

What the polar bias can and cannot do for cryptography: telling drift from eavesdropping in BB84, and certified quantum randomness

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Abstract

We ask where the qg description of a qubit, $qg_Z = \langle \sigma_z \rangle$ and its companions qg_X, qg_Y [1], is useful for cryptography. Post-quantum cryptography (lattice-based key encapsulation such as ML-KEM [2]) is classical and has no place for it. Quantum cryptography does, because its raw data are qg values. (i) In BB84 [3] over a qubit channel with relaxation, the sum $A_Z = qg(0) + qg(1)$ of the received polar biases is a T1 witness. Natural relaxation raises it, while intercept-resend adds symmetric errors that leave it unchanged. A likelihood-ratio monitor built on it detects intercept-resend as well as the standard QBER monitor, but raises no false alarms when the memory's T1 degrades, where the QBER monitor fires in 48–100% of windows. By construction it misses an attacker who mimics T1, which the QBER monitor catches, so the two are complementary and neither changes the secret-key rate. (ii) For a qubit random-number generator with a trusted measurement, the certified min-entropy is $H_{\min}(Z|E) = -\log_2 [(1 + \sqrt{1 - qg_X^2 - qg_Y^2})/2]$, independent of qg_Z . The usual output-bias estimate is unsafe whenever the state is mixed: it certifies 0.97 bits per shot where 0.015 are private. A qg estimate from $\pm$ -rotation test rounds is safe in every run, and a readout calibration recovers 10–50% more certified bits. All results are simulations, reproduced by scripts and pinned by regression tests.

1 Where qg does not apply: post-quantum cryptography

Post-quantum schemes such as ML-KEM (Kyber) [2] or lattice and hash-based signatures run on classical computers and are designed to resist quantum attacks. They involve no qubits and no measurement statistics, so a coordinate for qubit measurements has nothing to contribute. The threat they answer, Shor's algorithm, does not benefit from qg either; the project's reference notebook documents it as a non-application. We state this first so that the rest is not read as a claim about post-quantum security.

2 BB84: drift or eavesdropper?

Setting. Alice sends $|0\rangle, |1\rangle$ (Z basis) or $|\pm\rangle$ (X basis); the qubit waits in a memory or channel with amplitude damping γ (T1) and dephasing p ; Bob measures with an asymmetric detector ($e_{01} = 0.005, e_{10} = 0.01$). This fits matter-qubit links and on-chip demonstrations, not photon-polarization links, whose noise is mostly unital. Each sent state gives one qg value at Bob,

$$qg(0) = 1 - 2e(0 \rightarrow 1), \quad qg(1) = -1 + 2e(1 \rightarrow 0), \quad A_Z = qg(0) + qg(1) = 2[e(1 \rightarrow 0) - e(0 \rightarrow 1)], \quad (1)$$

and likewise in X. A_Z is the register-mean T1 witness of the companion paper [4]: relaxation creates only $1 \rightarrow 0$ errors and raises it, while intercept-resend on a fraction f of the qubits

Table 1: Alarm rates (1000 windows of $N = 2000$ compared bits). The last column is the Shor–Preskill secret fraction $1 - h(Q_Z) - h(Q_X)$ [5].

scenario	QBER monitor	qg monitor	secret fraction
baseline	0.007	0.006	0.720
T1 drift, γ 0.02 $\rightarrow$ 0.04	0.478	0.009	0.640
T1 drift, γ 0.02 $\rightarrow$ 0.06	0.962	0.006	0.567
intercept-resend, $f = 0.02$	0.218	0.271	0.668
intercept-resend, $f = 0.05$	0.861	0.823	0.594
intercept-resend, $f = 0.10$	1.000	0.994	0.481
T1-mimicking attack (extra $\gamma = 0.04$)	0.962	0.006	0.567

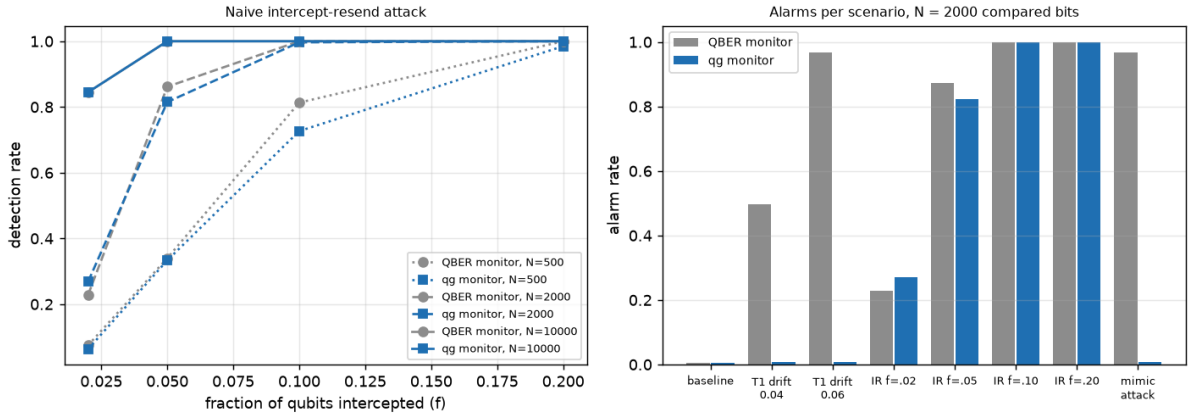


Figure 1: Left: detection of intercept-resend vs the intercepted fraction, for $N = 500$, 2000 and 10,000 compared bits. Right: alarm rates per scenario at $N = 2000$.

adds $f/4$ errors in every category and leaves it unchanged. The baseline channel has $\gamma_0 = 0.02$, $p_0 = 0.01$.

Monitors. Windows of N compared bits, 1% false alarms on the baseline. The *QBER monitor* tests the total error count. The *qg monitor* is a generalized likelihood ratio on the four error counts, in which “attack” ($f > 0$) must beat both “baseline” and “T1 drift” (γ free).

Results (Table 1, Fig. 1).

- *Equal detection, fewer false alarms.* The qg monitor detects intercept-resend as well as the QBER monitor at every N and f , but stays quiet when T1 degrades. The QBER monitor fires in 48–100% of those windows.
- *The price, by design.* An attacker who couples the qubit to her ancilla through an amplitude-damping interaction is, from Bob’s side, identical to T1 drift. The qg monitor misses her and the QBER monitor does not. Run together, QBER says that something changed and qg says what kind of change it is.
- *No security gain.* The secret fraction falls in the same way in all cases. A security proof must attribute every error to Eve [6]. The qg monitor is an operational diagnostic.

3 Certified random numbers

A qubit QRNG [7] prepares $|+\rangle$ and measures Z . Dephasing and thermal population leave the output unbiased, but then part of the randomness is classical noise that an adversary holding

Table 2: Certified bits per shot: mean estimate and, in parentheses, the fraction of runs above the truth (unsafe). 300 repetitions.

scenario	truth	naive	qg $\pm$	qg calibrated
ideal $ +\rangle$	1.000	0.965 (0)	0.512 (0)	0.681 (0)
T2, $V = 0.8$	0.322	0.964 (1.00)	0.245 (0)	0.286 (0)
T2, $V = 0.5$	0.100	0.965 (1.00)	0.077 (0)	0.087 (0)
T2, $V = 0.2$	0.015	0.965 (1.00)	0.009 (0)	0.010 (0)
thermal $p = 0.05$, $V = 0.9$	0.334	0.965 (1.00)	0.254 (0)	0.297 (0)
$e_{10} = 0.10$, $V = 0.95$	0.608	0.876 (1.00)	0.341 (0)	0.513 (0)

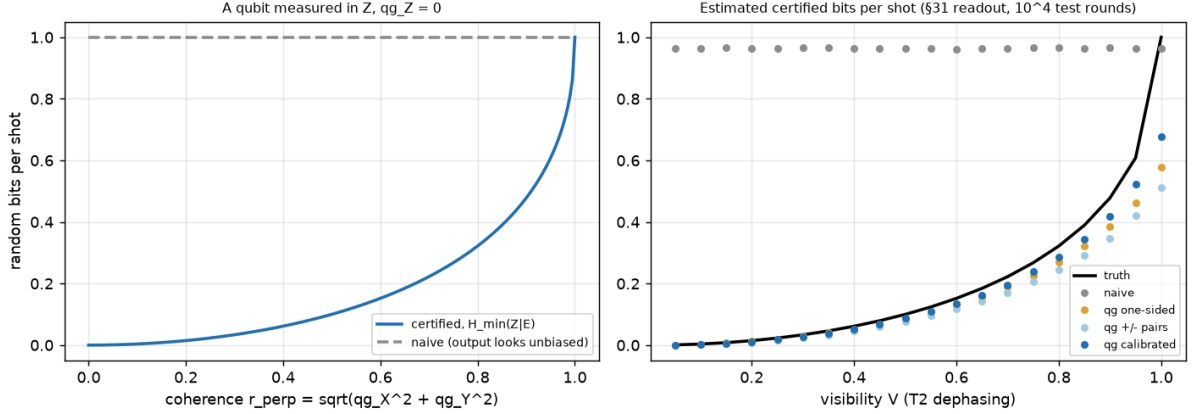


Figure 2: Left: certified bits per shot vs the coherence $r_{\perp}$ at $qg_Z = 0$, against the naive reading. Right: estimates vs the visibility V .

the environment can know. With a trusted measurement and an adversary holding the purification, her two conditional states are pure, with overlap $|\rho_{01}| = r_{\perp}/2$. By Helstrom [8] and the operational meaning of min-entropy [9],

$$P_{\text{guess}}(Z|E) = \frac{1 + \sqrt{1 - r_{\perp}^2}}{2}, \quad H_{\min}(Z|E) = -\log_2 P_{\text{guess}}, \quad r_{\perp} = \sqrt{qg_X^2 + qg_Y^2}. \quad (2)$$

The result is independent of qg_Z : only the coherence in the measured basis is private. For a pure state it reduces to the classical $-\log_2 \max(p_0, p_1)$. We checked Eq. (2) against an explicit purification.

Estimators. From 10^4 test rounds per setting, with the readout $qg_Z^{\text{meas}} = a + b qg_Z$ ($e_{01} = 0.015$, $e_{10} = 0.04$) calibrated by the heralded sweep of the companion paper. We compare four estimates. The *naive* one is the output-bias estimate $-\log_2 \max(\hat{p}_0, \hat{p}_1)$. The *qg one-sided* estimate takes $r_{\perp}$ from X and Y test rounds with a single rotation each. The *qg $\pm$* estimate measures each axis with both rotations, so the offset a cancels. The *qg calibrated* estimate is the $\pm$ one divided by the calibrated b . All qg estimates use a 3σ lower bound on $r_{\perp}$.

Results (Table 2, Fig. 2).

- *The output-bias estimate is unsafe* whenever the state is mixed: it overestimates in every run.
- *The qg estimate is safe in every run* when each axis is measured with both rotations. With a single rotation the readout offset inflates $r_{\perp}$, and at $V = 0.2$ the estimate exceeds the truth in 7% of runs.

- *Calibration pays.* Dividing by the calibrated b recovers 10–50% more certified bits.
- *A near-perfect source is shot-hungry to certify.* $H_{\min}$ has an infinite slope at $r_{\perp} = 1$, the qg pole, so the 3σ bound gives 0.68, 0.81 and 0.89 bits per shot with 10^4 , 10^5 and 10^6 test rounds.

4 Limitations

(i) Everything is a simulation of a single-qubit model with i.i.d. rounds. (ii) The QRNG is device-dependent (trusted measurement), not device-independent. (iii) Eq. (2) and the BB84 security bound are standard results; qg writes them in measured quantities. (iv) The BB84 monitor is a diagnostic: an attacker who mimics relaxation is invisible to it, and the key rate is unchanged. (v) The BB84 setting needs a relaxation mechanism (matter qubits); it does not describe photon-polarization links.

5 Code availability

`examples/bb84_qg_eve_vs_noise.py` and `examples/qrng_qg_certified.py` in <https://github.com/Viny2030/qang> (MIT license), with regression tests. Sections 40 and 41 of `RESEARCH_NOTES.md` give more detail, and `notebooks/qang_criptografia_colab.ipynb` reproduces fast versions in Colab.

Acknowledgments and use of AI tools

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